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Diploma in Engg. (Fourth Semester) Examination, Nov.-Dec. 2015

(Old Scheme)

(Electrical Engg. Branch)

ELECTRICAL MACHINES–I

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : *All questions are compulsory, unless mentioned otherwise.*

1. Choose the correct answer :

(i) Transformer action require a : 2

(a) Constant Magnetic Flux

(b) Increasing Magnetic Flux

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(c) Alternatic Magnetic Flux

(d) Decreasing Magnetic Flux

(ii) Efficiency of a Power Transformer is minimum at : 2

(a) Nearly full load

(b) 1/2 full load

(c) 1/4 full load

(d) 1/8 full load

(iii) A coil consist of : 2

(a) Two conductor

(b) Two coil side

(c) Two turn

(d) Four turn

(iv) The type of DC generator used for arc welding purpose is a : 2

(a) Series Generator

(b) Shunt Generator

(c) Cumulative Compound Generator

(d) Differential Compound Generator

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2. Explain the principle of transformer action. Also derive an expression for the EMF induced in a tranformer winding. 10

3. Draw the exact equivalent circuit of a transformer and describe briefly the various parameter involve in it. 10

Or

Explain why transformer rating is expressed in KVA or VA. Describe the significance of all the items mentioned on the name plate of a single phase transformer.

4. A 50 kVA, 2200/110 V, 50 Hz, transformer when tested :

Open circuit test (Low voltage side) : 400 W, 10 A, 110 V

Short circuit test (On high voltage side) : 808 W, 20.5 A, 90 V

Compute the parameter of the approximate aquivalent circuit. Also draw the exact equivalent circuit. 8

5. An industrial laod takes 100 A at 0.8 P.F. lag from a 3ϕ 11000/400 V, 50 Hz Star-Delta transformer, calculate : 8

(i) Power consume by load

(ii) kVA rating of transformer

(iii) Phase and line current on both H.V. and L.V. side.

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6. Give the constructional feature of D.C. machines, with a suitable diagram. Describe the constructional detail of a commutator also. 8

7. Derive the e.m.f. equation of DC generator. 8

8. Define the armature reactoin and commutation. 10

Or

A 10 kW, 250 V, 6 pole lap connected dc generator runs of 1200 rpm. Armature has 500 conductor for full load armature ohmic less are 200 W. Find flux/pole. Take 2 V as brush drop.

9. A 200 V shunt motor has $Ra = 0{\cdot}1\,\Omega$, Rf (Field Resistance) = $240\,\Omega$ and rotational loss 230 W on full load. The line current is 9·8 A with motor running at 1450 rpm. Determine : 10

(i) Machine power developed

(ii) Power output

(iii) Load torque

(iv) Full load efficiency

Or

Explain why the E.M.F. generated in the armature of a dc motor is called back E.M.F.

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10. Explain the speed control of DC Machine by Ward Leonard system. 10

11. Explain 3 point starter. 10

Or

Explain 4 point starter.

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